| | | |
|---|---|---|
| | | the voltmeter is always measuring the e.m.f. of the battery. |
| **22** | **A** | Between X and Y, the network is made of parallel resistors of resistance 70 Ω, 70 Ω, (40+20+80=140) Ω and (50+30+60=140) Ω.<br><br>By proportionality, the 1.5 A current is divided into 6 parts, i.e. 0.25 A.<br>Current of 0.50 A flows through each of the two 70 Ω resistors.<br>Currents each of 0.25 A flow through the (40+20+80=140) Ω and (50+30+60=140) Ω. Resistors respectively.<br><br>Hence, the current in 30 Ω resistor is 0.25 A. |
| **23** | **B** | The magnetic force provides the centripetal force necessary for the charge to move in a |